

Do your calculations on separate sheets. Write your results in the boxes, but justify your answer by summarizing your calculations and any necessary reasoning, so that we can evaluate how you arrived at the answers in the boxes. **No results without justification will be accepted.** For your justification, do it (quoting Einstein) "as simple as possible, but not simpler" ☺. Do not write in the margins.

3.5

PROBLEM 1

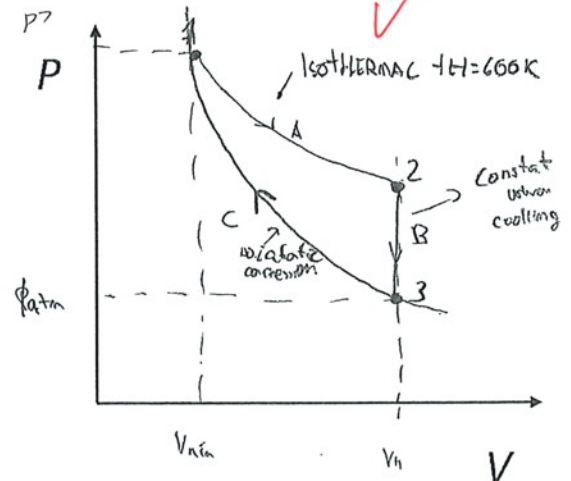
A gas is used as a heat engine operating between two thermal reservoirs at temperatures $T_H = 600 \text{ K}$ and $T_C = 300 \text{ K}$ according to the following cycle:

A – Isothermal expansion at T_H from a certain volume to a final volume of 500 liters.

B – Cooling at constant volume till the gas final pressure becomes 1 atm. to "atm"

C – Adiabatic compression (back to the initial state in A).

Assume that the gas behaves as a monatomic ideal gas. Use as universal gas constant, $R = 8.314 \text{ J K}^{-1} \text{ mol}^{-1}$.



1.1) Sketch the cycle in the PV diagram provided above. (0.5 point)

1.2) What is the gas' internal energy change in stage A, work in stage B, and heat in stage C? (1 point)

• We have that A is isothermal expansion $\Rightarrow \Delta U_A = 0$

• We know that $W = -P dV$ and how volume = constant at B $\Rightarrow W = 0$

• We know that in adiabatic process $Q = 0$

$$\Delta U_A (\text{J}) = 0$$

$$W_B (\text{J}) = 0$$

$$Q_C (\text{J}) = 0$$

1.3) Find the efficiency of this heat engine. (1 point)

$V_C = V_{\text{given}}$
 $V_D = V_{\text{ref}}$

For all cycle $\Delta U = 0 = W_C + Q_B \Rightarrow W_C = -Q_B$

From adiabatic: $\frac{V_H}{V_C} = \left(\frac{T_H}{T_C}\right)^{\frac{1}{\gamma-1}} = 2.828$

$$\eta = 0.2787$$

$$\eta = \frac{|W|}{|Q_H|} = 1 + \frac{\frac{3}{2}NR\Delta T}{\frac{3}{2}NR\ln\left(\frac{V_H}{V_C}\right)} = 1 - 0.7213$$

$$\eta = 0.2787$$

1.4) In one cycle, what is the entropy change of the gas ΔS and the global entropy change (gas plus heat reservoirs) ΔS_{global} . (1 point)

We have a reversible cycle $\Rightarrow \Delta S = 0$ because S is a state variable

How $\Delta S = 0 \Rightarrow \Delta S_{\text{global}} (\text{J/K}) = 0$ $\rightarrow$ Also this because if we consider reservoirs + gas as one only system

We have that $\Delta S = 0$ $\rightarrow$ decrease because $\Delta S = 0$ if it's one isolated system.

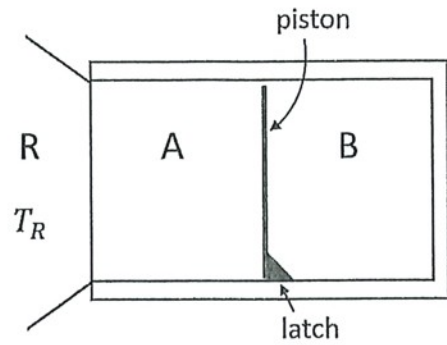
$$\Delta S (\text{J/K}) = 0$$

$$\Delta S_{\text{global}} (\text{J/K}) = 0$$

But entropy increases because the cycle is irreversible!

2.9**PROBLEM 2**

Ideal gases A and B are in a cylinder separated by a diathermic frictionless piston, as schematically shown in the figure. Gas A is also in thermal contact with a heat reservoir R at temperature $T_R = 300$ K. Initially, the pressure of A is higher than that of B (10 and 1 atm, respectively), and the piston is held by a latch, as shown. The gases occupy a total volume of 20 liters, and initially the piston is at the center of the cylinder. The volume of the latch is negligible.



2.1) Predict the final state of the system after the latch holding the piston is removed. Write the answers in the answer box for the final volumes V_f , temperature T_f and pressure P_f for the two gases. (1 point)

Initially we have $T_A = T_R = T_B = 300$ K. And how R will give Q all time and piston is diathermic $\Rightarrow T_A = T_B = T_R = 300$ K

With initial values we can compute that

$$n_A = 4,064 \text{ mols} \quad n_B = 0,406 \text{ mols} \quad \text{equilibrium}$$

P_{Af} (Pa)	$5,57 \times 10^5$
P_{Bf} (Pa)	$5,57 \times 10^5$
T_{Af} (K)	300
T_{Bf} (K)	300
V_{Af} (Pa)	18,18 L
V_{Bf} (Pa)	1,818 L

→ At finally we will have that $P_{Af} = P_{Bf}$

⇒ we compute all the cylinder we will have that

$$n_A + n_B \quad \frac{P_f = \frac{n_A R T_f}{V} = \frac{n_B R T_f}{V} = 5,57 \times 10^5 \text{ Pa} = P_{Af} = P_{Bf}$$

comparing both will have that:

$$V_{Af} = \frac{n_A R T_f}{P_f} = 18,18 \text{ L}$$

$$V_{Bf} = \frac{n_B R T_f}{P_f} = 1,818 \text{ L}$$

2.2) What is the heat exchanged with the reservoir in the process between the initial and the final states.

1. (0.5 point)

→ we have that for all the cylinder $\Delta U = Q_A + W_A + Q_B + W_B = 0$ → Because we are in internal state.

And they are ideal gas.

$$\text{How } \Delta U_{\text{system}} = \Delta U_R + \Delta U_{\text{cylinder}} = 0 \Rightarrow \Delta U_R = 0$$

And how

$$\Delta U = W + Q \quad \text{and } Q = 0 \Rightarrow Q_R = 0$$

2.3) What are the entropy changes for the two gases and for the reservoir? Is the process reversible or irreversible? (1 point)

$$\Delta S_A = \frac{Q_A}{T} = \frac{6056}{300} = 20,18 \text{ J/K}$$

$$Q_A = -W_A = +n_A R T \ln\left(\frac{V_f}{V_i}\right) = 6056,53 \text{ J}$$

$$Q_B = n_B R T \ln\left(\frac{V_f}{V_i}\right) = -1726,4 \text{ J} \Rightarrow \Delta S_B = \frac{Q_B}{T} = -5,754$$

How all system $\Delta S = 0$

$$\Rightarrow \Delta S_R = (20,18 - 5,754) = -14,426 \text{ J/K}$$

$$\Delta S_A \text{ (J/K)} = 20,18$$

$$\Delta S_B \text{ (J/K)} = -5,754$$

$$\Delta S_R \text{ (J/K)} = -14,426$$

(Reversible or irreversible?) (Circle)

Why? (Using less than 30 words)

reversible

we can apply an infinitesimal force on the piston until we arrive to initial position, and the diathermic walls will extract the heat and make process isothermal.

2.4) If we had allowed for reversible work interactions with an external work source, how much work would we have been able to extract from the system between the initial and the final states? (0.5 point)

$$W = -P_{ext} W_A + W_B = -P_{ext} \cdot n_A R T \cdot \ln\left(\frac{V_A}{V_0}\right) - n_B R T \cdot \ln\left(\frac{V_B}{V_0}\right)$$

$$= -6056.53 + 1726.4 = 4329.6 \text{ J}$$

$$W^{rev} (\text{J}) = 4329.6$$

$$+ 4330 \text{ J}$$

PROBLEM 3

I prepared iced coffee as follows: Three ice cubes from the freezer were placed in a glass at room temperature of 25°C . Then, hot coffee was poured directly on the ice cubes.

Assume that initially the ice cubes are at -15°C and the hot coffee at 85°C . The mass of the glass is 150 g, the volume of the coffee is 100 cm^3 , and the size of one ice cube is 50 cm^3 . Assume that coffee has approximately the same properties as water. Use the following property values (assumed to be temperature-independent): Heat capacity of glass = $0.8 \text{ J g}^{-1} \text{ K}^{-1}$. Heat capacity of water $4.184 \text{ J g}^{-1} \text{ K}^{-1}$. Heat capacity of ice: $2 \text{ J g}^{-1} \text{ K}^{-1}$. Density of water = 1.000 g cm^{-3} . Density of ice = 0.917 g cm^{-3} . Melting heat of ice = 334 J g^{-1} .

3.1) What is the increase in liquid volume between the initial and final states? (1 point)

→ we suppose that $T_f = 0^\circ\text{C}$, if it's different, we are going to find it while computing.
→ we are in the pressure and we have that: $\int dQ_{rev} = \Delta H = 0$. This is because not heat can be computed by the surroundings.

$$\Delta H = \Delta H_{ice} + \Delta H_{cup} + \Delta H_{coffee} = m_{ice} \cdot C_{P,ice} \cdot \Delta T + m \cdot \ln + m_{cup} \cdot C_{P,cup} \cdot \Delta T + m_{glass} \cdot C_{P,glass} \cdot \Delta T$$

(I have no time to put units in the

theory calculator
of my sheet
are.

$$\Delta V (\text{cm}^3) = -9.34$$

$$103$$

$$= 137.55 \cdot 2 \cdot 15 + m \cdot 334 \text{ J/g} + 100 \cdot 4.18 \cdot (-85) + 150 \cdot 0.8 \cdot (-25) = 0$$

$$\Rightarrow m = \frac{34432.5}{334} = 103.11 \text{ g} < \text{ice mass} \Rightarrow T_f = 0^\circ\text{C}$$

3.2) What is the final temperature of the coffee (once thermodynamic equilibrium is reached after pouring the coffee on the ice)? Assume that no water evaporates and that no heat is appreciably lost to the surroundings. (1 point)

$$\Delta V = V_f - V_0$$

$$m_{ice \text{ melted}} = 137.55 - 103.11$$

$$= 34.44 \text{ g}$$

$$V_0 = 100 + 150 = 250 \text{ cm}^3$$

$$V_f = 100 + 103.11 + 34.44 = 237.55 \text{ cm}^3$$

$$\Delta H = -9.34 \text{ cm}^3$$

$$V_{ice} = \frac{34.44}{0.917} = 37.56 \text{ cm}^3$$

$$V_{ice} = \frac{34.44}{0.917} = 37.56 \text{ cm}^3$$

$$T_f (\text{K}) = 273.15$$

→ As we compute before, the $T_f = 0^\circ\text{C}$, because not all ice turn in water.

3.3) Is the process reversible or irreversible, and why? (0.5 point)

Reversible or irreversible? (Circle)

Why? (Using less than 30 words)

It's irreversible because it's an spontaneous process. And can't be reversible making small changes in the surroundings. Not great but OK.

... because the heat interactions take place with finite differences of temperature.